EE2-08A MATHEMATICS

1. a) *Comments about the actual exam performance follow the solution, in italics. Where there are no comments, the questions was done well by most people.*

Show that the function

$$u(x,y) = \sinh x \cos y + 2\cosh x \sin y$$

satisfies Laplace's equation [4]

ii) Integrate the Cauchy-Riemann equations to find the conjugate function $v(x,y)$. [4]

iii) Show that $w = u + iv$ can be expressed as

$$w = C_1 \sin(C_2 z) + C_3$$

and determine the complex constants C_1, C_2, C_3 [5]

SOLUTION

i) First:

$$\begin{aligned} u_x &= \cosh x \cos y + 2\sinh x \sin y \\ u_{xx} &= \sinh x \cos y + 2\cosh x \sin y = u \end{aligned}$$

And

$$\begin{aligned} u_y &= -\sinh x \sin y + 2\cosh x \cos y \\ u_{yy} &= -\sinh x \cos y - 2\cosh x \sin y = -u \end{aligned}$$

Finally: $u_{xx} + u_{yy} = u - u = 0$, so it satisfies Laplace's equation.

Most problems occured with people not remembering how to differentiate cosh and sinh. Typical error:
$(\cosh x)' = i\sinh x, (\sinh x)' = -i\cosh x$ *and similar. Some replaced these with sin/cos but often got the identities (see solution iii) wrong. Others replaced them with exponentials but this led to very messy calculations, lots of errors.*

ii) Cauchy-Riemann equations:

$$\begin{aligned} u_x = v_y \quad &\Rightarrow \quad v = \int u_x \, dy \\ u_y = -v_x \quad &\Rightarrow \quad v = -\int u_y \, dx \end{aligned}$$

Solving the first one:

$$\begin{aligned} v &= \int u_x \, dy = \int (\cosh x \cos y + 2\sinh x \sin y) \, dy \\ &= \cosh x \sin y - 2\sinh x \cos y + f(x) \end{aligned}$$

And the second one:

$$\begin{aligned} v &= -\int u_y \, dx = -\int (-\sinh x \sin y + 2\cosh x \cos y) \, dx \\ &= \cosh x \sin y - 2\sinh x \cos y + g(y) \end{aligned}$$

Where f and g are both arbitrary functions.
Comparing both solutions we obtain that $f = g = constant$, hence

$$v = \cosh x \sin y - 2\sinh x \cos y + C, \qquad C \in \mathbb{R}$$

Many problems following from complicated/wrong answers in (i). Many forgot to include f,g, or just put constants. Only five people mentioned what f,g are: arbritrary functions.

iii) Using $\cosh x = \cos(ix)$ and $\sinh x = -i\sin(ix)$ we get:

$$\begin{aligned} w &= u + iv \\ &= -i\sin(ix)\cos y + 2\cos(ix)\sin y + i[\cos(ix)\sin y - 2(-i\sin(ix))\cos y] + iC \\ &= (2+i)\cos(ix)\sin y - (2+i)\sin(ix)\cos y + iC \\ &= (2+i)\sin(y - ix) + iC \\ &= (2+i)\sin(-iz) + iC \end{aligned}$$

Hence: $C_1 = 2+i$, $C_2 = -i$ and $C_3 = iC$ (with $C \in \mathbb{R}$).

Few followed this approach. Many people used the shortcut of looking at $w(x,0)$, but then there were many who had trouble with converting sinh *to* sin*, lots of incorrect manipulation to force the right form of answer. Others used the shortcut with $w(0,y)$ but often forgot a factor of i. If you let $x = 0$ you conclude that $z = iy$, hence $y = z/i$ is needed.*

b) The complex function $f(z) = \dfrac{e^{iz}}{z(z^2+9)}$ has simple poles at $z = \pm 3i$ and $z = 0$.

(i) Show that the residue at $z = 0$ is $1/9$; find the residue at $z = 3i$. [4]

Consider the contour integral $I = \oint_\Gamma \dfrac{e^{iz}}{z(z^2+9)}\, dz$,

where Γ is taken to be the union of a semi-circle of radius R, lying in the upper half-plane, with a small semi-circle of radius r indented into the lower half-plane, both centred at $z = 0$, and the real intervals $[-R, -r]$ and $[r, R]$.

(ii) Show that the contribution to I from the indented semi-circle of radius r, in the limit $r \to 0$, is $i\pi/9$. [5]

(iii) Show that the contribution to I from the arc of the larger semi-circle, in the limit $R \to \infty$, is zero. [3]

(iv) Hence use the Residue Theorem to show that

$$\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+9)}\, dx = \frac{\pi}{9}\left(1 - e^{-3}\right).$$

[5]

Recall that the residue of a complex function $F(z)$ at a pole $z = a$ of multiplicity m is given by the expression

$$\lim_{z \to a} \left\{ \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} \left[(z-a)^m F(z) \right] \right\}.$$

SOLUTION

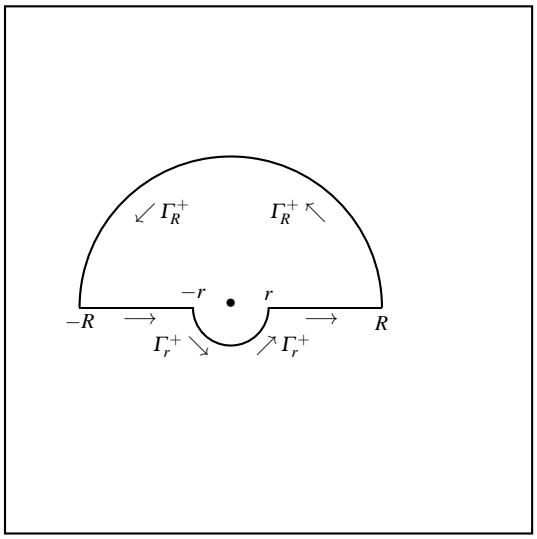
i) Residue at $z=0$:

$$\lim_{z\to 0} z\frac{e^{iz}}{z(z^2+9)} = \lim_{z\to 0}\frac{e^{iz}}{(z^2+9)} = \frac{1}{9}$$

Residue at $z=3i$:

$$\lim_{z\to 3i}(z-3i)\frac{e^{iz}}{z(z^2+9)} = \lim_{z\to 3i}\frac{e^{iz}}{z(z+3i)} = \frac{e^{3i^2}}{3i(6i)} = -\frac{e^{-3}}{18}$$

ii)



The contour Γ is shown above, with Γ_r: small semi-circle, Γ_R: big semi-circle. $\Gamma = \Gamma_R \cup [-R,-r] \cup \Gamma_r \cup [r,R]$., and superscript $^+$ indicating positive, counterclockwise orientation.

For the small semicircle we have

$$I = \int_{\Gamma_r} f(z)\, dz$$

Let $z = re^{i\theta}$, where $\pi \le \theta \le 2\pi$ - we are moving anticlockwise from $(-r,0)$ to $(r,0)$. Then $dz = ire^{i\theta} = izd\theta$. Substituting:

$$I = \int_{\pi}^{2\pi} \frac{e^{i(re^{i\theta})} i \not{z}\, d\theta}{\not{z}\,(r^2 e^{i2\theta}+9)}$$

$$\lim_{r\to 0} I = \frac{i}{9}\int_{\pi}^{2\pi} 1\, d\theta = \frac{i\pi}{9}$$

As $re^{i\theta}$ and $r^2e^{i2\theta} \to 0$ as $r \to 0$.

Many people seemed to think that one can apply Cauchy's Residue theorem: obtain the residue at zero and multiply by $2\pi i$. Then, as we're integrating around half of a circle, just multiply that by 1/2.

This does give the correct answer but does **not** *follow from Cauchy's theorem, but instead is a distinct result not covered in lectures. Many others had problems with the substitution for z and the limits needed. Lots of unwarranted manipulations to force the answer into the required form.*

iii) Since:

- the only singularities of the function are poles
- $m = 1 > 0$
- $|F(z)| = \dfrac{1}{z(z^2+9)} \to 0$ fast enough as $R \to \infty$

$\lim_{R\to\infty} \int_{\Gamma_R} f(z)\, dz = 0$ as conditions hold for Jordan's lemma.

iv) Using the Residue Theorem:

$$2\pi i \left(\frac{1}{9} - \frac{e^{-3}}{18}\right) = \oint_{\Gamma} f(z)\, dz = \int_{\Gamma_R} f(z)\, dz + \int_{-R}^{-r} f(z)\, dz + \int_{\Gamma_r} f(z)\, dz + \int_{r}^{R} f(z)\, dz$$

Taking the limits as $r \to 0$ and $R \to \infty$

$$2\pi i \left(\frac{1}{9} - \frac{e^{-3}}{18}\right) = \frac{i\pi}{9} + 0 + \int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2+9)} dx$$

$$i\left(\frac{\pi}{9} - \frac{\pi e^{-3}}{9}\right) = \int_{-\infty}^{\infty} \frac{\cos x}{x(x^2+9)} dx + i \int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+9)} dx$$

The integral with the cosine term is zero since the integrand is an odd function. This leaves:

$$\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+9)} dx = \frac{\pi}{9}\left(1 - e^{-3}\right)$$

Many different versions of the integrals, taking the limits $r \to 0, R \to \infty$ at the wrong time, not including the term from (ii), forgetting about cosine. Mostly sloppy errors. Plenty of incorrect manipulations to force the required answer.

2. a) Derive the formula for the Laplace transform of the Heaviside function:

$$\mathscr{L}\,[H(t-a)] = \frac{e^{-as}}{s}. \qquad [4]$$

SOLUTION

$$\mathscr{L}\,[H(t-a)] = \int_0^\infty H(t-a)e^{-st}\,dt = \int_a^\infty e^{-st}\,dt$$
$$= \frac{e^{-st}}{-s}\bigg|_a^\infty = \frac{e^{-as}}{s}$$

Since $\lim_{t\to\infty} \frac{e^{-st}}{-s} = 0$, provided $\Re(s) > 0$.

Only about 1 in 7 stated the condition for convergence as $t \to \infty$. Many used a circular argument, saying it is a result of a shift theorem.

b) Hence, or otherwise, obtain the Laplace transform of the rectangular pulse fuction:

$$f(t) = \begin{cases} 3 & \text{for} \quad 0 \le t \le 2 \\ 0 & \text{for} \quad t > 2 \end{cases} \qquad [3]$$

SOLUTION

$$f(t) = 3\,[H(t) - H(t-2)]$$
$$\mathscr{L}\,[f(t)] = \frac{3}{s} - \frac{3e^{-2s}}{s}$$

Many thought the transform is $1/s$ for $0 \le t < 2$ and zero for $t \ge 2$.

c) Use partial fractions to obtain the inverse Laplace transform:

$$\mathscr{L}^{-1}\left[\frac{1}{s\,(s^2+4s+3)}\right]. \qquad [5]$$

SOLUTION

$$\frac{1}{s(s^2+4s+3)} = \frac{1}{s(s+3)(s+1)} = \frac{A}{s} + \frac{B}{s+3} + \frac{C}{s+1}$$
$$1 = A(s+3)(s+1) + Bs(s+1) + Cs(s+3)$$

Let:

$$s = 0 \quad \Rightarrow \quad 1 = 3A \quad \Rightarrow \quad A = \frac{1}{3}$$
$$s = -1 \quad \Rightarrow \quad 1 = -2C \quad \Rightarrow \quad C = -\frac{1}{2}$$
$$s = -3 \quad \Rightarrow \quad 1 = 6B \quad \Rightarrow \quad B = \frac{1}{6}$$

Hence

$$\mathscr{L}^{-1}\left[\frac{1}{s\,(s^2+4s+3)}\right] = \frac{1}{3}\mathscr{L}^{-1}\left[\frac{1}{s}\right] + \frac{1}{6}\mathscr{L}^{-1}\left[\frac{1}{s+3}\right] - \frac{1}{2}\mathscr{L}^{-1}\left[\frac{1}{s+1}\right]$$
$$= \frac{1}{3} + \frac{1}{6}e^{-3t} - \frac{1}{2}e^{-t}$$

d) Use (b) and (c) and Laplace transforms to show that the differential equation

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 3x = \begin{cases} 3 & \text{for} \quad 0 \le t \le 2 \\ 0 & \text{for} \quad t > 2 \end{cases},$$

which satifies the initial conditions $x=1$ and $\dfrac{dx}{dt}=0$ when $t=0$, has solution

$$x(t)=\begin{cases} 1 & \text{for} \quad 0\le t\le 2 \\ -\frac{1}{2}e^{-3(t-2)}+\frac{3}{2}e^{-(t-2)} & \text{for} \quad t>2 \end{cases} \qquad [\,8\,]$$

SOLUTION

$$\mathscr{L}\,[\ddot{x}+4\dot{x}+3x]=\mathscr{L}\,[f(t)]$$

$$s^2F(s)-x(0)s-x'(0)+4\,[sF(s)-x(0)]+3F(s)=\frac{3}{s}-\frac{3e^{-2s}}{s}$$

Using the initial conditions $x(0)=1$ and $x'(0)=0$ we have:

$$\begin{aligned}(s^2+4s+3)F(s)&=\frac{3}{s}+s+4-\frac{3e^{-2s}}{s}\\ &=\frac{s^2+4s+3}{s}-\frac{3e^{-2s}}{s}\\ F(s)&=\frac{1}{s}-\frac{3e^{-2s}}{s(s^2+4s+3)}\end{aligned}$$

$$x(t)=\mathscr{L}^{-1}\,[F(s)]=1-3\mathscr{L}^{-1}\left[\frac{e^{-2s}}{s(s^2+4s+3)}\right]$$

(now use second shift theorem:)

$$\begin{aligned}&=1-3H(t-2)\mathscr{L}^{-1}\left[\frac{1}{s(s^2+4s+3)}\right]_{t\to t-2}\\ &=1-3H(t-2)\left[\frac{1}{3}+\frac{1}{6}e^{-3t}-\frac{1}{2}e^{-t}\right]_{t\to t-2}\\ &=1+H(t-2)\left[-1-\frac{1}{2}e^{-3(t-2)}+\frac{3}{2}e^{-(t-2)}\right]\\ &=\begin{cases}1 & 0\le t<2\\ -\dfrac{1}{2}e^{-3(t-2)}+\dfrac{3}{2}e^{t-2} & t\ge 2\end{cases}\end{aligned}$$

As required.

Most people did not see the cancellation in the last line of finding $F(s)$, many just took $\dfrac{3}{s}+s+4$ and divided by s^2+4s+3, resulting in much more work, lots of scope for arithmetic error. Problems also arising from including the wrong initial condition or none at all. Many people solved the problem twice: once for $0\le t<2$, where $\ddot{x}+4\dot{x}+3x=3$, correctly obtaining the solution $x=1$. But these people then continued: for $t\ge 2$, we have $\ddot{x}+4\dot{x}+3=0$ with the **same** *initial conditions, which is not automatic and needs arguing: the initial conditions were for $t=0$, not for $t=2$! Solving the new ODE gives $-\dfrac{1}{2}e^{-3t}+\dfrac{3}{2}e^{t}$ and shifting $t\to t-2$ to give the required answer is also not automatic and needs careful argument. Again, lots of incorrect manipulation to force results into the required form.*